

Artificial Intelligence for Cybersecurity

Applications, Benefits, Limitations, New Risks and Perspectives

Research question

How does artificial intelligence transform the detection and prevention of cyber threats, and what are the limitations and new risks associated with its use in cybersecurity?

Author Judicaël OUEDRAOGO

Date September 29, 2026

Nature Documentary and analytical research report

Prepared according to principles of scientific traceability, methodological caution and reproducibility.

Abstract

This study investigates the use of artificial intelligence (AI) in cybersecurity, with particular attention to intrusion detection, malware detection, phishing detection, behavioral analysis, Security Operations Center (SOC) support, SIEM operations and threat intelligence. It simultaneously examines limitations and risks associated with AI-based security systems, including data quality, class imbalance, false positives and false negatives, explainability, generalization, bias, adversarial machine learning, data poisoning, privacy and model security.

The study follows a documentary and analytical approach rather than an original laboratory experiment. Quantitative results are presented as study- and protocol-specific findings, not as universal performance claims. For each reported result, the study, year, dataset, task, model, metric and limitation are identified when available. This distinction is essential because high benchmark performance does not by itself demonstrate equivalent effectiveness in a different operational environment. NIST and ENISA publications likewise emphasize risk management, robustness, traceability, adversarial threats and protection across the AI lifecycle [4, 5, 12, 14].

The central conclusion is organized around two complementary dimensions: **AI for Security**, meaning the use of AI to strengthen cybersecurity, and **Security for AI**, meaning the protection of AI data, models, infrastructure and uses.

Keywords

Artificial intelligence; cybersecurity; machine learning; deep learning; intrusion detection; malware detection; phishing detection; behavioral analysis; adversarial machine learning; SOC; SIEM; threat intelligence.

Contents

Abstract	i
Keywords	i
1 Introduction and Research Framework	1
1.1 Context	1
1.2 Research Problem	1
1.3 Objectives	1
1.3.1 General objective	1
1.3.2 Specific objectives	2
1.4 Research Questions	2
1.5 Scope and Method	2
2 Foundations of AI and Cybersecurity	3
2.1 Artificial Intelligence	3
2.2 Machine Learning	3
2.2.1 Supervised Learning	3
2.2.2 Unsupervised Learning and Anomaly Detection	3
2.2.3 Classification and Anomaly Detection	3
2.3 Deep Learning	3
2.4 Generative AI	4
2.5 Core Cybersecurity Concepts	4
3 Applications of AI in Cybersecurity	5
3.1 Intrusion Detection	5
3.2 Malware Detection	5
3.3 Phishing Detection	5
3.4 Behavioral Analysis	5
3.5 SOC and SIEM	6
3.6 Threat Intelligence	6
4 Benefits of AI in Cybersecurity	7
4.1 Automation	7

4.2	Scalability	7
4.3	Detection Speed	7
4.4	Pattern Recognition	7
4.5	Anomaly Detection	7
4.6	Alert Prioritization	7
4.7	Continuous Monitoring and Analyst Assistance	8
5	Limitations and Risks	9
5.1	False Positives	9
5.2	False Negatives	9
5.3	Data Quality	9
5.4	Generalization	9
5.5	Explainability	9
5.6	Bias	10
5.7	Adversarial Machine Learning	10
5.7.1	Evasion	10
5.7.2	Poisoning	10
5.7.3	Privacy Attacks and Model Manipulation	10
5.8	Privacy	10
5.9	Model Security	10
6	AI Used by Attackers	12
6.1	A Dual-Use Capability	12
6.2	Automated Phishing and Social Engineering	12
6.3	Reconnaissance Assistance	12
6.4	Malware and Content Generation Assistance	12
6.5	Deepfakes and Impersonation	12
6.6	Defensive Adaptation	13
7	Quantitative Evidence	14
7.1	Interpretation Principle	14
7.2	Metric Comparison	16
7.3	Scientific Reading of Results	16
8	Case Studies and Research Trends	19
8.1	Classical Machine Learning	19
8.2	Deep Learning	19
8.3	Hybrid Architectures	19
8.4	Behavioral Detection	19
8.5	AI-Assisted SOC	19
8.6	Adversarial Robustness	20

8.7	Explainable AI	20
8.8	Privacy-Preserving Approaches	20
8.9	AI Governance	20
8.10	Observed Evidence versus Future Hypotheses	20
9	Conditions for Trustworthy Deployment	21
9.1	Data Governance	21
9.2	Multi-Dimensional Evaluation	21
9.3	Model Monitoring	21
9.4	Human Validation	21
9.5	Pipeline Security	21
9.6	Documentation and Reproducibility	21
10	Cross-Cutting Analysis	22
10.1	AI Primarily Transforms Analytical Capacity	22
10.2	The Benchmark Is Not the Field	22
10.3	The Metric Depends on Risk	22
10.4	AI Creates a Dependency Loop	22
10.5	A Socio-Technical Approach	22
11	Perspectives	23
11.1	Robust AI Security	23
11.2	AI-Assisted SOC	23
11.3	Human-AI Collaboration	23
11.4	Explainable AI for Cybersecurity	23
11.5	Federated and Privacy-Preserving Learning	23
11.6	Continuous Monitoring and Drift	23
11.7	Better Datasets	24
11.8	Cross-Dataset Evaluation	24
11.9	Governance and Standards	24
11.10	AI for Security + Security for AI	24
12	Conclusion	25
	Limitations of this Study	25
	Appendix – Quantitative Traceability Note	26
	References	27

List of Figures

2.1	Conceptual workflow for AI-assisted security detection.	4
5.1	Conceptual taxonomy of risks surrounding AI-based cybersecurity.	11
6.1	Conceptual relationship between AI-enabled offensive scaling and defensive adaptation.	13
7.1	Reported accuracy – study-specific results, not directly comparable across datasets.	16
7.2	Reported precision, recall and F1-score where available. Missing values are shown as unavailable.	17
7.3	Distribution of quantitative study entries by application domain in the selected corpus.	17
7.4	Number of quantitative study entries by publication year in the selected corpus. This is not a global scientific trend.	18

List of Tables

3.1 Comparison of major AI applications in cybersecurity. 6

7.1 Study comparison and traceability. 15

7.2 Reported metrics across the selected corpus. Missing values are not treated as zero. 16

CHAPTER 1

Introduction and Research Framework

1.1 Context

Digital transformation has expanded the number of connected systems, cloud services, applications, Internet of Things devices and data flows. This expansion also increases the volume of technical events generated by information systems: logins, authentication events, network flows, process activity, files, application logs and security alerts. A modern SOC must triage, correlate and interpret large quantities of information under time pressure.

The challenge is not only volume. Threats evolve, change their characteristics and may resemble legitimate behavior. Signature- and rule-based controls remain useful for many known cases, while machine learning can learn statistical regularities from security data and search for deviations. Research reviews nevertheless show that performance depends strongly on data, features, metrics and evaluation protocols [1, 10].

ENISA's Threat Landscape 2025 analyzed 4,875 incidents observed between July 1, 2024 and June 30, 2025, illustrating the operational scale at which security teams must prioritize events [6]. The broader security environment therefore creates a strong motivation for automation, but automation must remain measurable and controlled.

1.2 Research Problem

How does artificial intelligence transform the detection and prevention of cyber threats, and what are the limitations and new risks associated with its use in cybersecurity?

The question requires benefits and risks to be examined together. Accuracy alone is insufficient: a security system must also be robust, maintainable, appropriately explainable, adapted to its context and resilient to changing data distributions.

1.3 Objectives

1.3.1 General objective

Analyze how artificial intelligence transforms cyber-threat detection and prevention while assessing its technical limitations, operational risks and future prospects.

1.3.2 Specific objectives

- Identify major applications of AI in defensive cybersecurity.
- Compare documented approaches and quantitative results.
- Examine data quality, evaluation metrics and generalization.
- Analyze adversarial machine learning, data poisoning and attacks against AI systems.
- Study offensive uses of AI from a defensive risk perspective.
- Identify emerging research directions and deployment challenges.

1.4 Research Questions

- What cybersecurity functions are being transformed by AI?
- What benefits are documented?
- What technical and operational limitations remain?
- What new risks arise from AI itself?
- How is AI changing offensive cyber capabilities?
- What conditions are required for trustworthy deployment?

1.5 Scope and Method

This is a documentary and analytical study, not an original laboratory experiment. The quantitative corpus is a selected set of documented studies. Results are retained with their datasets, models, metrics and limitations whenever available. No missing statistic, DOI or experimental detail is inferred.

CHAPTER 2

Foundations of AI and Cybersecurity

2.1 Artificial Intelligence

Artificial intelligence refers to methods that enable computational systems to produce predictions, classifications, recommendations or generated content from defined objectives. In cybersecurity, AI is mainly used to process security data, learn complex relationships and assist human decision-making.

2.2 Machine Learning

Machine learning (ML) consists of methods that learn a model from data. A cybersecurity model may distinguish benign and malicious events, classify malware families or identify unusual behavior.

2.2.1 Supervised Learning

The model learns from labeled examples. A classifier may use observations labeled as normal or malicious. Its strength is direct evaluation on held-out data; its limitation is dependence on label quality and representativeness.

2.2.2 Unsupervised Learning and Anomaly Detection

Unsupervised methods search for structure without requiring exhaustive labels. An autoencoder, for example, may learn to reconstruct behavior regarded as normal; reconstruction error can then become an anomaly signal.

2.2.3 Classification and Anomaly Detection

Classification asks which class an event belongs to. Anomaly detection asks whether an event sufficiently deviates from learned behavior. Hybrid systems can combine both approaches.

2.3 Deep Learning

Deep learning uses multilayer neural networks to learn representations. CNNs, recurrent architectures, LSTMs and hybrid architectures can capture local patterns, sequential dependencies

and complex representations. A 2025 IEEE Access survey of deep-learning intrusion detection research highlights persistent challenges involving data quality, scalability, adversarial robustness, privacy and interpretability [1].

2.4 Generative AI

Generative AI produces content such as text or code. Defensively, it can assist incident summarization, document retrieval and investigation support. It also creates risks such as hallucinated information, leakage of sensitive data and offensive misuse. NIST's generative-AI profile provides a dedicated risk-management framework [2].

2.5 Core Cybersecurity Concepts

IDS

Intrusion Detection System: monitors events to identify suspicious or malicious activity.

SOC

Security Operations Center: people, processes and technologies used to monitor, analyze and respond to security events.

SIEM

Security Information and Event Management: centralizes and correlates logs and security events.

Threat intelligence

collection, qualification and contextualization of threat information to support security decisions.

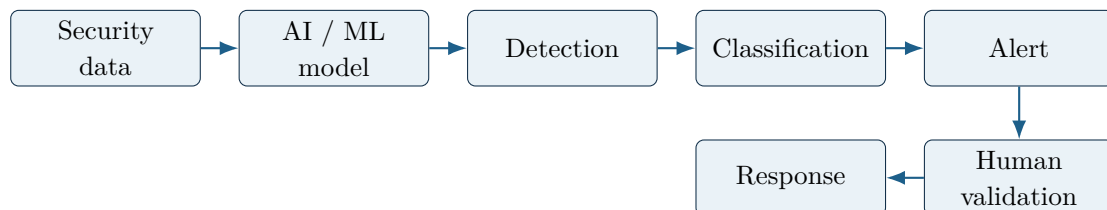


Figure 2.1: Conceptual workflow for AI-assisted security detection.

Source: Author's synthesis of ML, IDS and SOC concepts.

CHAPTER 3

Applications of AI in Cybersecurity

3.1 Intrusion Detection

ML and deep-learning models can analyze network flows, packets, system events and logs. Supervised models can classify known attack categories, while anomaly-detection methods can identify deviations from learned normal behavior. Telikani et al. combine an autoencoder with a cost-sensitive multitask SVM and report 92.2% recall, 96.2% precision and 94.3% F1-score across their evaluation on UNSW-NB15 and BoT-IoT [13]. These values remain specific to the study protocol and datasets.

3.2 Malware Detection

Malware detection can use static, dynamic or hybrid analysis. Static analysis examines file or program characteristics without necessarily executing the sample; dynamic analysis observes behavior during execution. A hybrid approach combines information from both sources. The TUANDROMD study by Soni et al. compares KNN, SVM and Decision Tree and reports accuracies of 97.65%, 98.62% and 99.10%, respectively [9]. These results do not constitute a universal ranking of malware-detection algorithms; they describe one dataset and experimental setup.

3.3 Phishing Detection

Phishing detection can use URLs, lexical characteristics, webpage structure and behavioral signals. A classifier can learn statistical differences between legitimate and malicious websites. A 2024 UBMK study using the UCI Phishing Websites dataset reports, for Random Forest, 96.7% accuracy, 96.1% precision, 98.2% recall and 97.1% F1-score [3]. The conclusion must remain limited to that protocol.

3.4 Behavioral Analysis

Behavioral analysis models sequences or activity profiles to identify significant deviations. It can concern users, sessions, processes, connections or resource interactions. Mohan and Kumar report 98.8% accuracy and 98.7% F1-score for an SVM plus Voting Classifier using Random Forest and Logistic Regression on CICIDS2017 [7]. A 2026 study combining 1D-CNN, Bi-LSTM

and an autoencoder reports 98.1% accuracy, 97.9% precision, 98.3% recall and 98.1% F1-score [11]. The precise dataset is marked unavailable in this report because it was not sufficiently established in the metadata reviewed.

3.5 SOC and SIEM

In a SOC, AI can assist alert correlation, deduplication, clustering of similar incidents, classification and prioritization. In a SIEM, it can help discover relationships across large volumes of logs and reduce repetitive work. This assistance does not eliminate analysts. High-impact decisions should retain appropriate human validation, particularly when evidence is ambiguous or the cost of error is high.

3.6 Threat Intelligence

AI can assist information extraction from reports, indicator classification, signal clustering and relationship discovery. It can accelerate the transformation of heterogeneous information into actionable context, but outputs must be verified because incorrect extraction or correlation can propagate false conclusions.

Table 3.1: Comparison of major AI applications in cybersecurity.

Application	Typical data	AI technique	Main objective	Main limitation
Intrusion detection	Network flows, logs	Classification, anomaly detection	Detect suspicious activity	Dataset and distribution shift
Malware detection	Files, features, behavior	ML/DL classifiers	Classify malicious software	Label quality and generalization
Phishing detection	URLs, webpages, lexical features	Classification	Identify fraudulent sites	Campaign evolution
Behavioral analysis	User/process/session activity	Anomaly detection, sequence models	Identify deviations	Baseline drift
SOC/SIEM	Alerts, logs, incidents	Correlation, ranking, LLM assistance	Reduce analyst workload	False positives and automation risk
Threat intelligence	Reports, indicators, text	Extraction, classification, correlation	Prioritize threat information	Verification and context dependence

CHAPTER 4

Benefits of AI in Cybersecurity

4.1 Automation

AI can automate repetitive processing such as classification, enrichment, clustering and initial triage. Automation can reduce manual workload when controls are carefully validated.

4.2 Scalability

Models can process large numbers of events more consistently than a human analyst could inspect individually. Scalability, however, depends on computing resources, data pipelines and monitoring.

4.3 Detection Speed

Automated inference can produce signals rapidly after data ingestion. Speed is valuable when response time matters, but rapid incorrect decisions can also increase operational risk.

4.4 Pattern Recognition

ML can identify combinations of features that are difficult to encode as simple rules. This can be useful for complex or high-dimensional security data.

4.5 Anomaly Detection

Anomaly detection can identify behavior that differs from a learned baseline, which is potentially useful when exact signatures are unavailable. The challenge is distinguishing malicious anomalies from legitimate changes.

4.6 Alert Prioritization

AI can rank or group alerts using contextual information, potentially helping analysts focus on events requiring attention. The ranking itself must be evaluated because prioritization errors can hide important incidents.

4.7 Continuous Monitoring and Analyst Assistance

AI supports continuous processing of security telemetry and can summarize evidence, suggest relationships or accelerate searches. The most defensible model is generally human-AI augmentation rather than universal analyst replacement.

CHAPTER 5

Limitations and Risks

5.1 False Positives

False positives generate investigation workload, alert fatigue and possible unnecessary intervention. A technically accurate classifier can still be operationally unsuitable if its false-positive burden is too high.

5.2 False Negatives

False negatives represent missed malicious activity. Their importance depends on the threat, the asset and the consequences of compromise; recall and other risk-sensitive metrics are therefore essential.

5.3 Data Quality

Security datasets may be incomplete, noisy, outdated, imbalanced or biased toward particular environments. Class imbalance is especially important because rare attacks can be overwhelmed by benign examples.

5.4 Generalization

High benchmark performance does not automatically imply real-world performance. Changes in networks, software, attacker behavior, sensors and data distributions can reduce effectiveness. Cross-dataset and time-based evaluation are therefore important.

5.5 Explainability

Complex models may be difficult to interpret. In security operations, analysts may need to understand why an alert was produced, especially when automated actions have significant consequences.

5.6 Bias

Training data and model design can introduce systematic differences in performance across environments or classes. Bias should be assessed as an empirical property rather than assumed away.

5.7 Adversarial Machine Learning

Adversarial machine learning studies attacks and mitigations involving AI systems. NIST's 2025 taxonomy distinguishes attack and mitigation concepts including evasion and poisoning [14].

5.7.1 Evasion

An adversary may attempt to cause a deployed model to misclassify an input. Defensively, this motivates robustness testing and monitoring.

5.7.2 Poisoning

Poisoning involves manipulating training or learning data so that model behavior is degraded or altered. The risk is particularly relevant when training pipelines ingest data from sources that cannot be fully trusted.

5.7.3 Privacy Attacks and Model Manipulation

AI systems can also create privacy risks through unintended information exposure, and models may be manipulated through weaknesses in their surrounding pipeline. These issues make model security part of cybersecurity rather than an optional add-on [8].

5.8 Privacy

Cybersecurity telemetry can contain identifiers, communications metadata, credentials, internal architecture information and other sensitive material. AI processing therefore requires appropriate access controls, minimization, retention and governance.

5.9 Model Security

Models, datasets, feature pipelines, prompts, inference interfaces and supporting infrastructure become security assets. Protecting them is necessary to preserve the integrity of the defensive system.

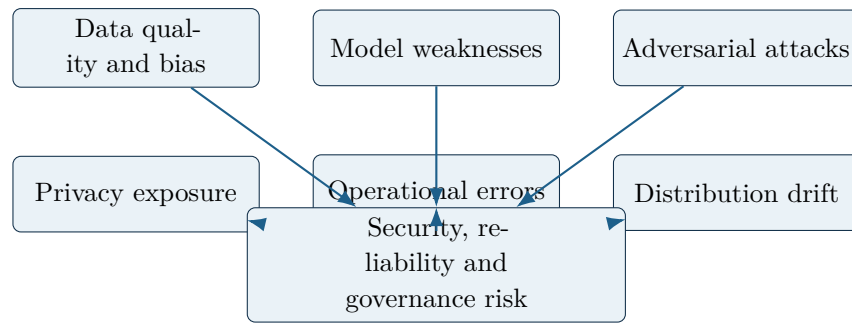


Figure 5.1: Conceptual taxonomy of risks surrounding AI-based cybersecurity.

Source: Author's synthesis based on NIST and ENISA risk concepts.

CHAPTER 6

AI Used by Attackers

6.1 A Dual-Use Capability

AI is dual-use. The same capabilities that help defenders process information can help attackers scale information processing, content creation and adaptation. This chapter remains analytical and defensive.

6.2 Automated Phishing and Social Engineering

Generative systems can reduce the effort required to produce convincing multilingual messages and tailored content. This increases the potential volume of social-engineering attempts and raises the importance of verification and awareness controls.

6.3 Reconnaissance Assistance

AI can assist the organization and summarization of publicly available information. From a defensive perspective, organizations should reduce unnecessary exposure and monitor for suspicious activity rather than rely on obscurity.

6.4 Malware and Content Generation Assistance

AI may lower barriers to producing or adapting code and malicious content. This report does not provide operational attack procedures or malicious code; the relevant defensive issue is increased potential for scaling and adaptation.

6.5 Deepfakes and Impersonation

Synthetic audio, images and video can support impersonation and fraud. Defensive controls increasingly require identity verification mechanisms that do not rely on a single communication channel or media format.

6.6 Defensive Adaptation

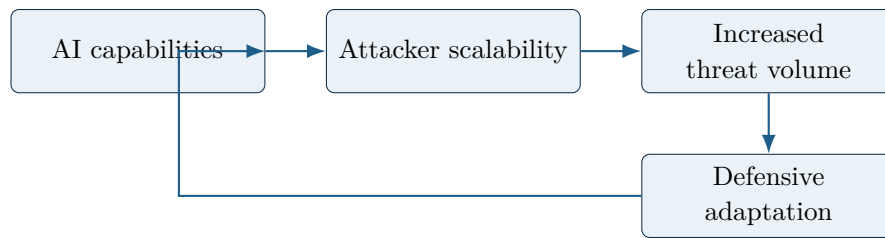


Figure 6.1: Conceptual relationship between AI-enabled offensive scaling and defensive adaptation.

Source: Conceptual synthesis; no operational attack procedure is provided.

CHAPTER 7

Quantitative Evidence

7.1 Interpretation Principle

The corpus contains six quantitative study entries. Each result must be read together with its dataset, task, model, metric and protocol. Accuracy, precision, recall, F1-score and MCC measure different properties and should not be treated as interchangeable.

Table 7.1: Study comparison and traceability.

ID	Year	Domain	Dataset	Model / method	Metrics / main result	Limitation
1	2024	Intrusion detection	UNSW-NB15 + BoT-IoT	Autoencoder + cost-sensitive multitask SVM	Recall 92.2%; Precision 96.2%; F1 94.3%	Dataset/protocol-specific; not universal IDS performance.
2	2025	Android malware	TUANDROMD	KNN; SVM; Decision Tree	Accuracy 97.65%; 98.62%; 99.10%	Specific experimental setup; not a universal algorithm ranking.
3	2024	Phishing	UCI Phishing Websites	Random Forest	Accuracy 96.7%; Precision 96.1%; Recall 98.2%; F1 97.1%	Dataset and protocol specific.
4	2025	Behavioral IDS	CICIDS2017	SVM + Voting Classifier (RF + LR)	Accuracy 98.8%; F1 98.7%	Benchmark-specific generalization.
5	2026	Behavioral anomaly detection	N/A	1D-CNN + Bi-LSTM + Autoencoder	Accuracy 98.1%; Precision 97.9%; Recall 98.3%; F1 98.1%	Dataset unavailable in reviewed metadata; real-world validation and computational cost require further study.
6	2024	Unseen / zero-day attack detection	CIC-MalMem-2022	Random Forest + Autoencoder	Accuracy 99.9892%; Precision 100%; Recall 99.9803%; F1 99.9901%; MCC 99.8313%	Unseen-attack definition and dataset are specific; does not prove universal zero-day detection.
6b	2024	Unseen / zero-day attack detection	CIC-MalMem-2022	XGBoost + Autoencoder	Accuracy 99.9741%; Precision 100%; Recall 99.9533%; F1 99.9976%; MCC 99.8002%	Same protocol limitation.

7.2 Metric Comparison

Table 7.2: Reported metrics across the selected corpus. Missing values are not treated as zero.

Study / model	Accuracy	Precision	Recall	F1	MCC
Telikani et al.	–	96.2%	92.2%	94.3%	–
TUANDROMD — KNN	97.65%	–	–	–	–
TUANDROMD — SVM	98.62%	–	–	–	–
TUANDROMD — Decision Tree	99.10%	–	–	–	–
Phishing — Random Forest	96.7%	96.1%	98.2%	97.1%	–
CICIDS2017 — Voting	98.8%	–	–	98.7%	–
Behavioral — hybrid	98.1%	97.9%	98.3%	98.1%	–
CIC-MalMem — RF+AE	99.9892%	100%	99.9803%	99.9901%	99.8313%
CIC-MalMem — XGB+AE	99.9741%	100%	99.9533%	99.9976%	99.8002%

7.3 Scientific Reading of Results

The selected studies show that very high values can be obtained on specific benchmarks. They do not establish that one model is globally superior, nor that a model will retain the same performance after deployment. Dataset construction, class balance, train-test separation, feature engineering and attack definitions can materially influence the reported values. In particular, the CIC-MalMem results should not be interpreted as proof that AI detects all zero-day attacks.

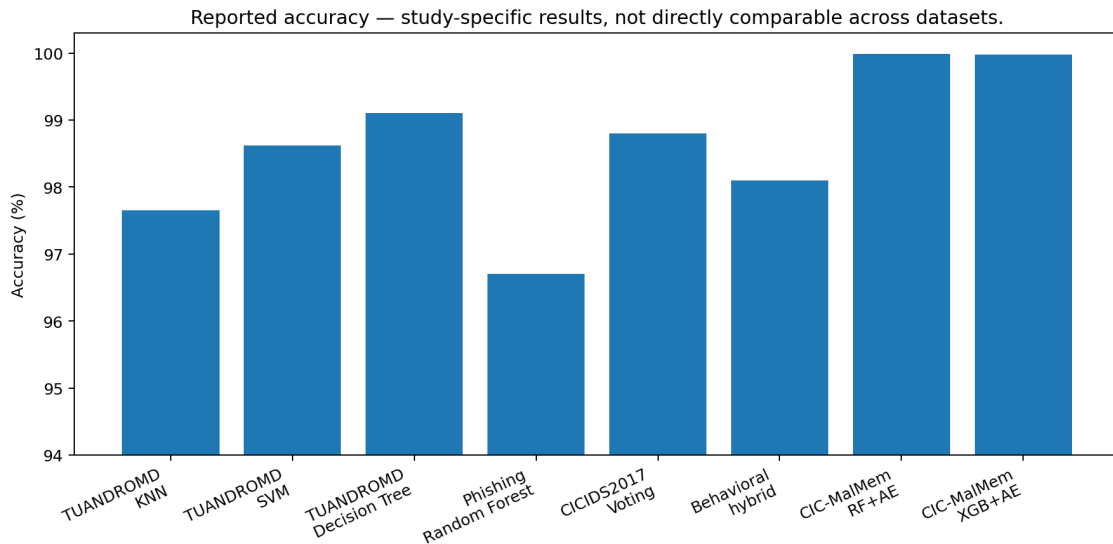


Figure 7.1: Reported accuracy – study-specific results, not directly comparable across datasets.

Source: Author's plot from the selected quantitative corpus.

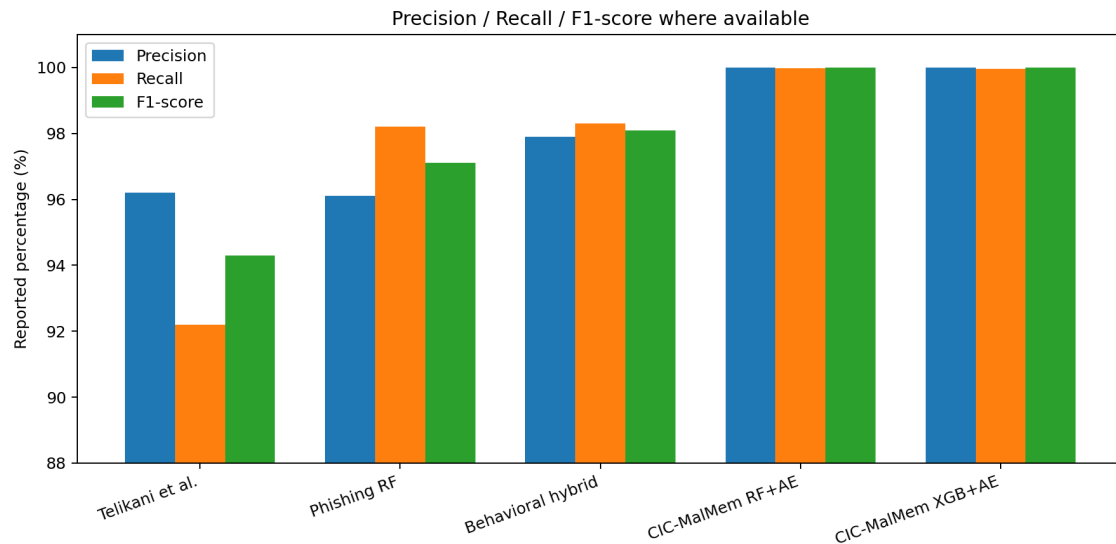


Figure 7.2: Reported precision, recall and F1-score where available. Missing values are shown as unavailable.

Source: Author's plot from the selected quantitative corpus.

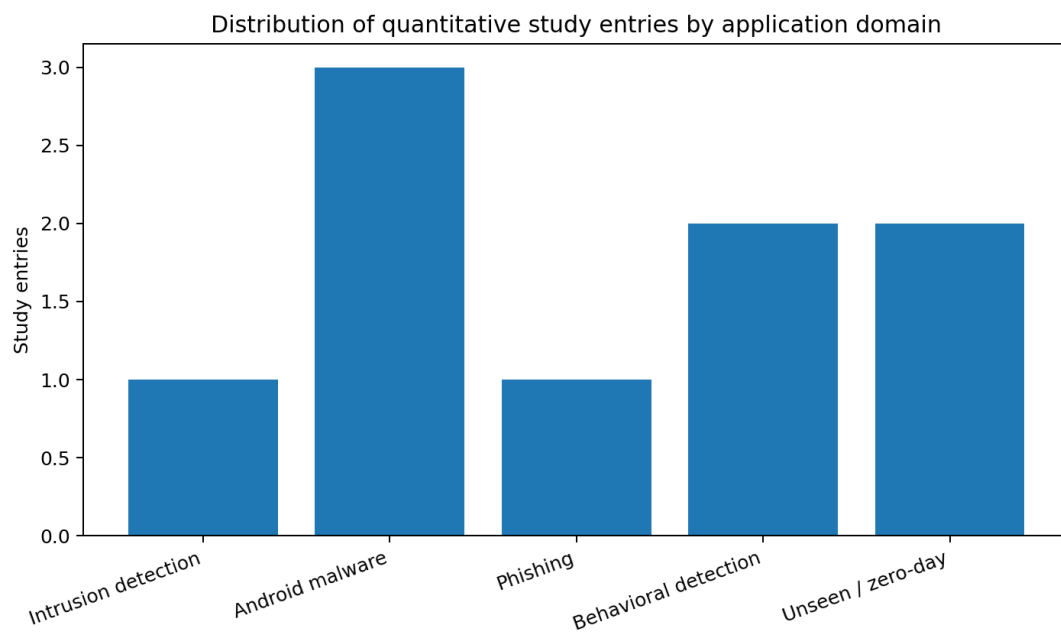


Figure 7.3: Distribution of quantitative study entries by application domain in the selected corpus.

Source: Author's classification of the selected corpus; not a global literature census.

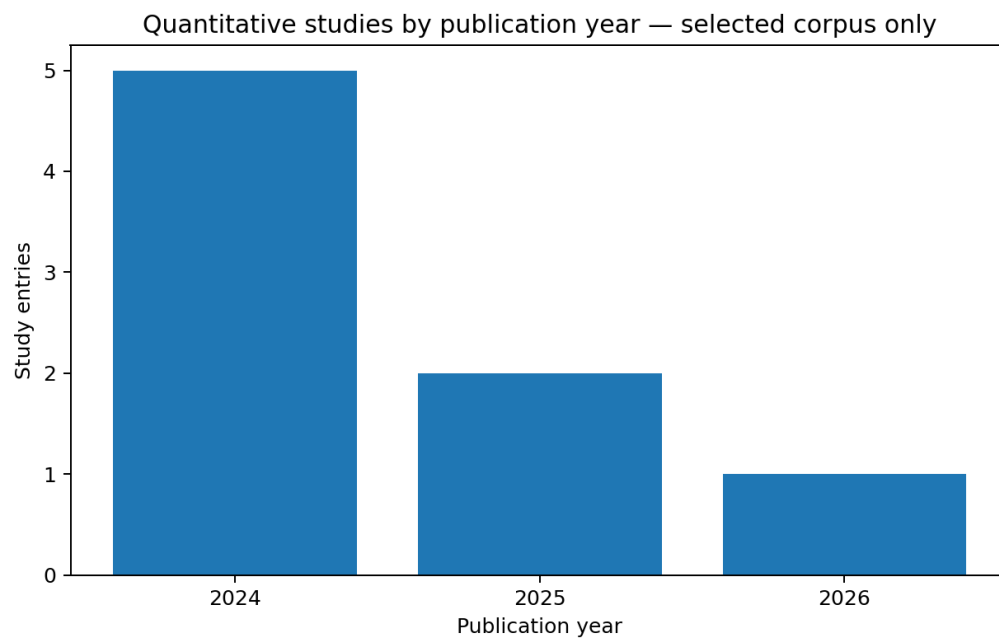


Figure 7.4: Number of quantitative study entries by publication year in the selected corpus. This is not a global scientific trend.

Source: Author's count of the selected corpus.

CHAPTER 8

Case Studies and Research Trends

8.1 Classical Machine Learning

Tree methods, SVMs, k-nearest neighbors and linear models remain relevant because they can work well on engineered security features and may be easier to evaluate and operate than very large architectures.

8.2 Deep Learning

Deep neural networks are increasingly studied for network intrusion detection, malware classification and behavioral analysis. Their representation-learning capacity is attractive for complex data, but computational cost, interpretability and robustness remain important considerations.

8.3 Hybrid Architectures

Hybrid models combine complementary mechanisms such as autoencoders with classifiers, or CNNs with recurrent models. The rationale is to combine representation learning, anomaly detection and supervised classification. The benefit must still be demonstrated under a clearly defined evaluation protocol.

8.4 Behavioral Detection

Behavioral approaches move beyond fixed signatures by modeling activity over time. This is promising for identifying deviations, but legitimate behavior also changes, creating a need for continuous baseline management.

8.5 AI-Assisted SOC

AI-assisted SOC research focuses on alert triage, correlation, investigation support and natural-language interfaces. A useful deployment must integrate AI with existing controls, analyst workflows and auditability rather than treating the model as an isolated component.

8.6 Adversarial Robustness

Adversarial machine learning has become a distinct security research area. Robustness should be evaluated against plausible threat models and across the full data and model pipeline [14].

8.7 Explainable AI

Explainability can help analysts understand evidence and challenge model outputs. The required degree of explanation depends on the operational context and consequence of errors.

8.8 Privacy-Preserving Approaches

Federated learning, data minimization and privacy-preserving techniques may reduce the need to centralize sensitive telemetry. They also introduce engineering and governance trade-offs.

8.9 AI Governance

Governance concerns risk ownership, documentation, monitoring, access control, incident response and accountability. NIST AI RMF provides a general framework for managing AI risks [12].

8.10 Observed Evidence versus Future Hypotheses

Observed evidence refers to results documented by the selected studies or authoritative organizations. **Future research hypotheses** include expected benefits from cross-dataset evaluation, robust model monitoring, privacy-preserving collaboration and improved human-AI interfaces. These are directions for investigation, not established universal outcomes.

CHAPTER 9

Conditions for Trustworthy Deployment

9.1 Data Governance

Datasets should be documented, versioned, quality-checked and governed according to sensitivity and provenance. Labels and sampling procedures should be recorded.

9.2 Multi-Dimensional Evaluation

Evaluation should include appropriate metrics, confusion matrices, robustness checks, computational cost and, when relevant, operational measures such as alert burden and detection latency.

9.3 Model Monitoring

Deployed models require monitoring for drift, performance degradation, unusual inputs and pipeline failures. Retraining should be governed rather than automatic by default.

9.4 Human Validation

Human oversight is important when evidence is uncertain or decisions have significant consequences. Analysts should be able to review and challenge automated recommendations.

9.5 Pipeline Security

Data ingestion, feature generation, training, model storage, inference interfaces and access controls must be protected as a single security pipeline.

9.6 Documentation and Reproducibility

A trustworthy system should document data sources, preprocessing, model versions, hyperparameters where appropriate, evaluation protocols and limitations. Reproducibility improves scientific and operational accountability.

CHAPTER 10

Cross-Cutting Analysis

10.1 AI Primarily Transforms Analytical Capacity

The strongest common effect across applications is the ability to process and structure large quantities of information. AI can therefore be understood primarily as an analytical capacity multiplier.

10.2 The Benchmark Is Not the Field

A benchmark is a controlled measurement environment. A production network has changing assets, users, software, attacks and data quality. Generalization must therefore be tested rather than assumed.

10.3 The Metric Depends on Risk

A metric becomes meaningful only in relation to the operational objective. A false negative can be more costly than a false positive in some contexts, while excessive false positives can make a system unusable in others.

10.4 AI Creates a Dependency Loop

Once security teams depend on AI, the availability and integrity of the AI pipeline become part of the security problem. A compromised or degraded model can therefore affect the defensive system itself.

10.5 A Socio-Technical Approach

Cybersecurity AI combines algorithms, people, processes, data, governance and infrastructure. Technical performance is necessary but not sufficient for trustworthy deployment.

CHAPTER 11

Perspectives

11.1 Robust AI Security

Future systems should be designed and tested for robustness against manipulation, distribution shift and unexpected inputs.

11.2 AI-Assisted SOC

Future SOC platforms may integrate correlation, summarization, investigation assistance and prioritization while preserving audit trails and human control.

11.3 Human-AI Collaboration

The practical objective is to allocate tasks according to comparative strengths: machines for large-scale processing and humans for contextual judgment, accountability and ambiguous cases.

11.4 Explainable AI for Cybersecurity

Explainability research should focus on explanations that are useful to analysts rather than explanations that are merely technically attractive.

11.5 Federated and Privacy-Preserving Learning

Collaborative learning may allow multiple organizations to benefit from shared patterns without centralizing all raw telemetry. Its privacy and security guarantees must nevertheless be validated.

11.6 Continuous Monitoring and Drift

Model behavior should be monitored over time because data distributions and attacker behavior evolve.

11.7 Better Datasets

Future research requires more representative, current and well-documented cybersecurity datasets, including realistic temporal splits and diverse environments.

11.8 Cross-Dataset Evaluation

Testing across datasets can reveal whether a model learned generalizable security patterns or dataset-specific artifacts.

11.9 Governance and Standards

Standards and governance frameworks can help organizations define responsibilities, risk acceptance, documentation and monitoring requirements.

11.10 AI for Security + Security for AI

This report identifies this dual perspective as a central direction for cybersecurity. **AI for Security** concerns intrusion detection, malware detection, phishing detection, behavioral analytics, SOC assistance and threat intelligence. **Security for AI** concerns data protection, model integrity, adversarial robustness, privacy, access control, monitoring and governance. A mature cybersecurity strategy needs both dimensions.

CHAPTER 12

Conclusion

Artificial intelligence is transforming cybersecurity by increasing the ability to process large volumes of security data, identify patterns, detect anomalies and support analysts. The selected research corpus demonstrates that ML, deep learning and hybrid architectures can produce strong quantitative results on specific datasets and experimental protocols.

However, AI does not eliminate cybersecurity problems. Its effectiveness depends on data quality, model design, evaluation methodology, generalization, robustness, explainability, human oversight and secure deployment. High benchmark performance should therefore be interpreted as evidence about a particular experimental setting rather than as proof of universal real-world effectiveness.

At the same time, AI creates a new attack surface. Data, models, pipelines, interfaces and AI-assisted workflows must themselves be protected. Offensive uses can increase the scale and adaptability of phishing, social engineering, content generation and other cyber activities, reinforcing the need for defensive adaptation.

The answer to the research question is therefore twofold: AI changes cybersecurity primarily by augmenting the capacity to analyze and prioritize complex security information, while its limitations and new risks require disciplined data governance, robust evaluation, human oversight and security of the AI system itself. Future cybersecurity must consequently develop both **AI for Security** and **Security for AI**.

Limitations of this Study

The report is documentary and uses a selected quantitative corpus rather than a systematic meta-analysis of the entire literature. The studies use different datasets, protocols and definitions, so their metrics are not directly comparable. One 2026 study in the corpus does not provide sufficiently established dataset metadata in the reviewed material and is therefore marked N/A. These limitations are part of the traceability of the report rather than information to be concealed.

Appendix – Quantitative Traceability Note

For every quantitative result retained in this report, the intended traceability fields are: study, year, dataset, task, model, metric, reported result, limitation and source. Missing information is represented as N/A or –. No missing values are replaced by zero, and no benchmark result is generalized beyond its documented experimental context.

References

- [1] F. M. Anis, M. Alabdullatif, S. Aljbli, and M. Hammoudeh, “A survey on the applications of deep learning in network intrusion detection systems to enhance network security,” *IEEE Access*, vol. 13, pp. 185 357–185 373, 2025. DOI: [10.1109/ACCESS.2025.3624952](https://doi.org/10.1109/ACCESS.2025.3624952). [Online]. Available: <https://doi.org/10.1109/ACCESS.2025.3624952>.
- [2] C. Autio et al., “Artificial intelligence risk management framework: Generative artificial intelligence profile,” National Institute of Standards and Technology, Tech. Rep. NIST AI 600-1, 2024. DOI: [10.6028/NIST.AI.600-1](https://doi.org/10.6028/NIST.AI.600-1). [Online]. Available: <https://doi.org/10.6028/NIST.AI.600-1>.
- [3] A. Ayyar et al., “Comparative analysis and practical implementation of machine learning algorithms for phishing website detection,” in *2024 9th International Conference on Computer Science and Engineering (UBMK)*, 2024. DOI: [10.1109/UBMK63289.2024.10773479](https://doi.org/10.1109/UBMK63289.2024.10773479). [Online]. Available: <https://doi.org/10.1109/UBMK63289.2024.10773479>.
- [4] European Union Agency for Cybersecurity, “Artificial intelligence and cybersecurity research,” ENISA, Tech. Rep., 2023. [Online]. Available: <https://www.enisa.europa.eu/publications/artificial-intelligence-and-cybersecurity-research>.
- [5] European Union Agency for Cybersecurity, “Cybersecurity of ai and standardisation,” ENISA, Tech. Rep., 2023. DOI: [10.2824/277479](https://doi.org/10.2824/277479). [Online]. Available: <https://doi.org/10.2824/277479>.
- [6] European Union Agency for Cybersecurity, “Enisa threat landscape 2025,” ENISA, Tech. Rep., 2025. [Online]. Available: <https://www.enisa.europa.eu/publications/enisa-threat-landscape-2025>.
- [7] J. Mohan and M. R. Kumar, “Intelligent intrusion detection system combining behavioral analysis and machine learning methods,” in *2025 3rd International Conference on Data Science and Network Security (ICDSNS)*, 2025. DOI: [10.1109/ICDSNS65743.2025.11168668](https://doi.org/10.1109/ICDSNS65743.2025.11168668). [Online]. Available: <https://doi.org/10.1109/ICDSNS65743.2025.11168668>.
- [8] A. Oprea, A. Singhal, and A. Vassilev, “Poisoning attacks against machine learning: Can machine learning be trustworthy?” *Computer*, vol. 55, no. 11, pp. 94–99, 2022. DOI: [10.1109/MC.2022.3190787](https://doi.org/10.1109/MC.2022.3190787). [Online]. Available: <https://doi.org/10.1109/MC.2022.3190787>.
- [9] H. Shah, V. Shah, N. Soni, V. Vadhavana, and K. Patel, “A comparative analysis for android malware detection using machine learning models,” in *2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*, 2025. DOI: [10.1109/ICMCSI64620.2025.10883385](https://doi.org/10.1109/ICMCSI64620.2025.10883385). [Online]. Available: <https://doi.org/10.1109/ICMCSI64620.2025.10883385>.

-
- [10] K. Shaukat, S. Luo, V. Varadharajan, I. A. Hameed, and M. Xu, “A survey on machine learning techniques for cyber security in the last decade,” *IEEE Access*, vol. 8, pp. 222 310–222 354, 2020. DOI: [10.1109/ACCESS.2020.3041951](https://doi.org/10.1109/ACCESS.2020.3041951). [Online]. Available: <https://doi.org/10.1109/ACCESS.2020.3041951>.
 - [11] V. Singh, A. Parvez, R. Krishnamoorthy, H. Devanna, K. Sridevi, and V. B. Gujar, “An intelligent behavioral anomaly detection framework for next-generation digital forensics using hybrid deep learning and context-aware feature fusion,” in *2026 4th International Conference on Artificial Intelligence and Machine Learning Applications Theme: Healthcare and Internet of Things (AIMLA)*, 2026. DOI: [10.1109/AIMLA67915.2026.11522297](https://doi.org/10.1109/AIMLA67915.2026.11522297). [Online]. Available: <https://doi.org/10.1109/AIMLA67915.2026.11522297>.
 - [12] E. Tabassi, “Artificial intelligence risk management framework (ai rmf 1.0),” National Institute of Standards and Technology, Tech. Rep. NIST AI 100-1, 2023. DOI: [10.6028/NIST.AI.100-1](https://doi.org/10.6028/NIST.AI.100-1). [Online]. Available: <https://doi.org/10.6028/NIST.AI.100-1>.
 - [13] A. Telikani et al., “A cost-sensitive machine learning model with multitask learning for intrusion detection in iot,” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 3880–3890, 2024. DOI: [10.1109/TII.2023.3314208](https://doi.org/10.1109/TII.2023.3314208). [Online]. Available: <https://doi.org/10.1109/TII.2023.3314208>.
 - [14] A. Vassilev, A. Oprea, A. Fordyce, H. Anderson, X. Davies, and M. Hamin, “Adversarial machine learning: A taxonomy and terminology of attacks and mitigations,” National Institute of Standards and Technology, Tech. Rep. NIST AI 100-2e2025, 2025. DOI: [10.6028/NIST.AI.100-2e2025](https://doi.org/10.6028/NIST.AI.100-2e2025). [Online]. Available: <https://doi.org/10.6028/NIST.AI.100-2e2025>.